

Enterprise Parcel Management System

Courier & Parcel Services | B.Tech DBMS Project

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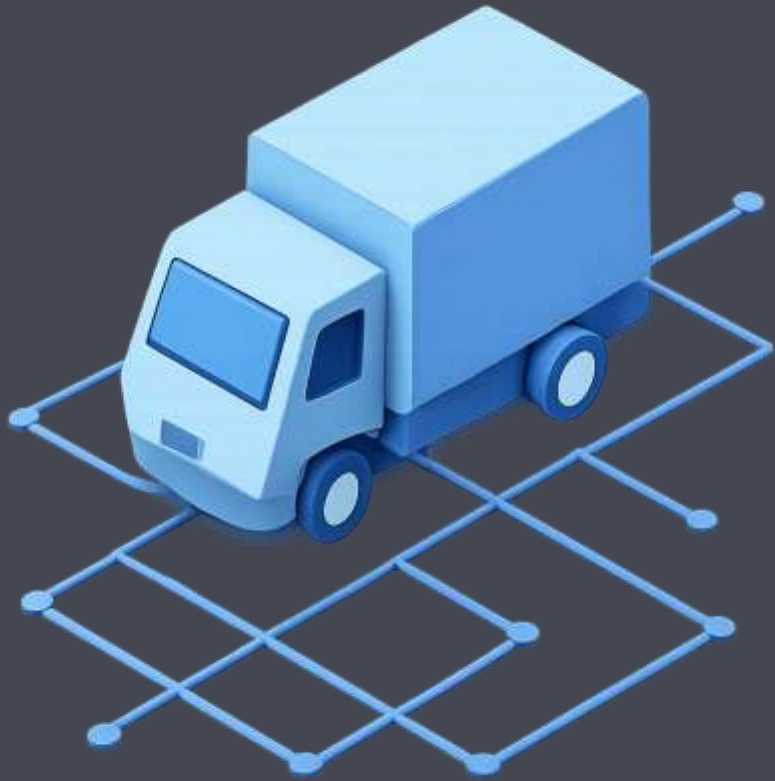
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Introduction

Critical to Modern Logistics

Courier and parcel management underpins the global supply chain, connecting businesses and consumers at every level.

Manual Systems Fall Short

Paper-based and phone-driven operations lead to inefficiencies, delays, and significant customer dissatisfaction.

Automation is the Answer

Digitization enables real-time tracking, streamlined workflows, and true operational excellence for courier enterprises.

⚠️ PROBLEM STATEMENT

The Challenges We Face



Manual Handling

Paper-based records are prone to errors, misplacement, and irreversible data loss.

Tracking Issues

Customers lack real-time visibility into parcel location and delivery status.

Data Redundancy

Duplicate information scattered across multiple disconnected systems.

Delayed Deliveries

Inefficient routing and poor resource allocation cause persistent delivery delays.

Project Objectives



Automate Operations

End-to-end parcel operations and booking fully digitized and automated.



Minimize Errors

Eliminate data inconsistencies through centralized, validated database management.



Real-Time Tracking

Live status updates and location visibility for every parcel in the system.



Enhance Satisfaction

Boost customer experience and overall operational efficiency across all branches.

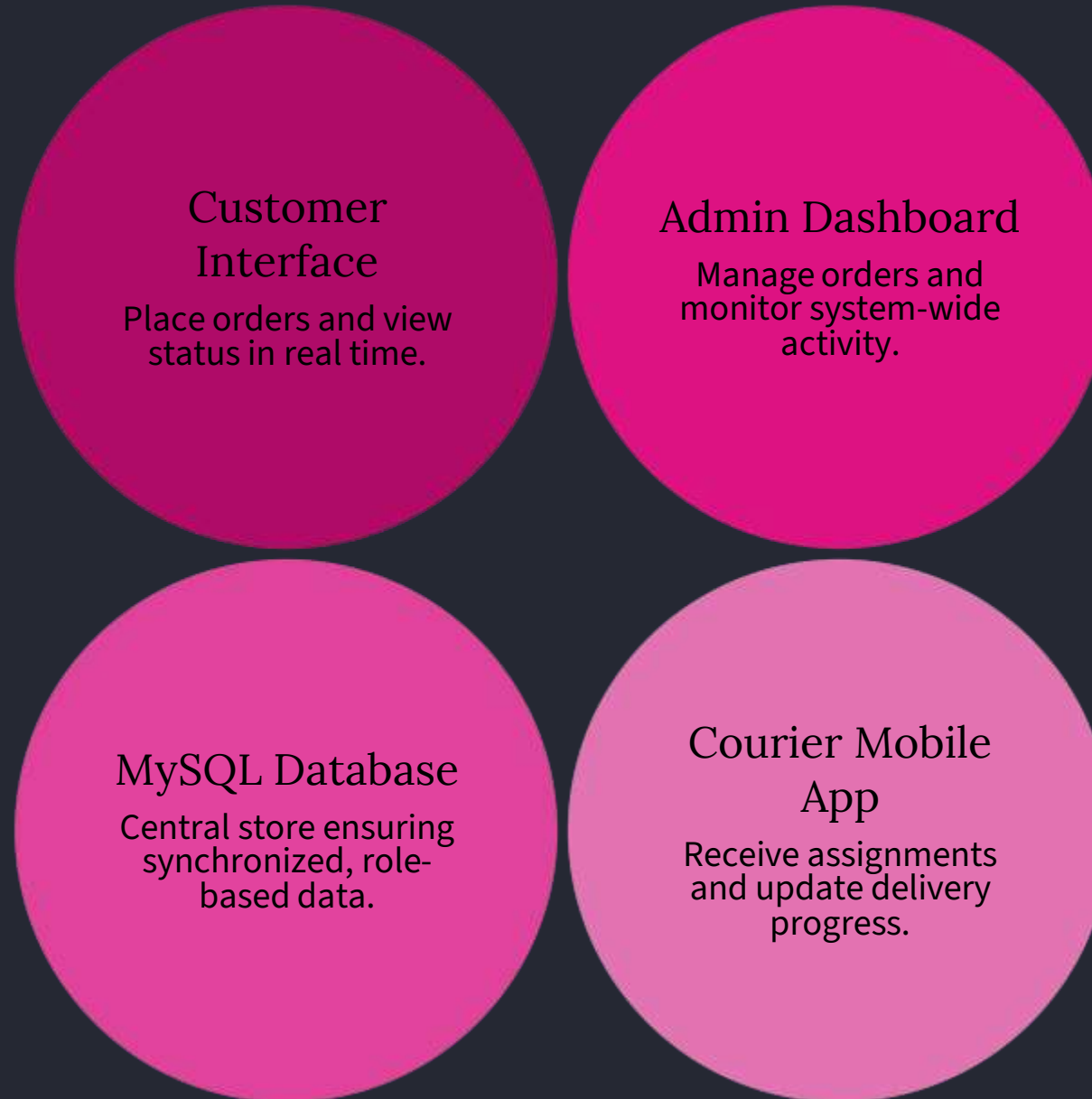
Existing vs. Proposed System

Feature		Existing System	Proposed System
Booking		Manual	Automated
Tracking		Phone-based	Real-time Digital
Records		Paper	Database
Errors		High	Minimal
Speed		Slow	Fast



The proposed system addresses every critical failure point of the legacy manual approach.

System Architecture



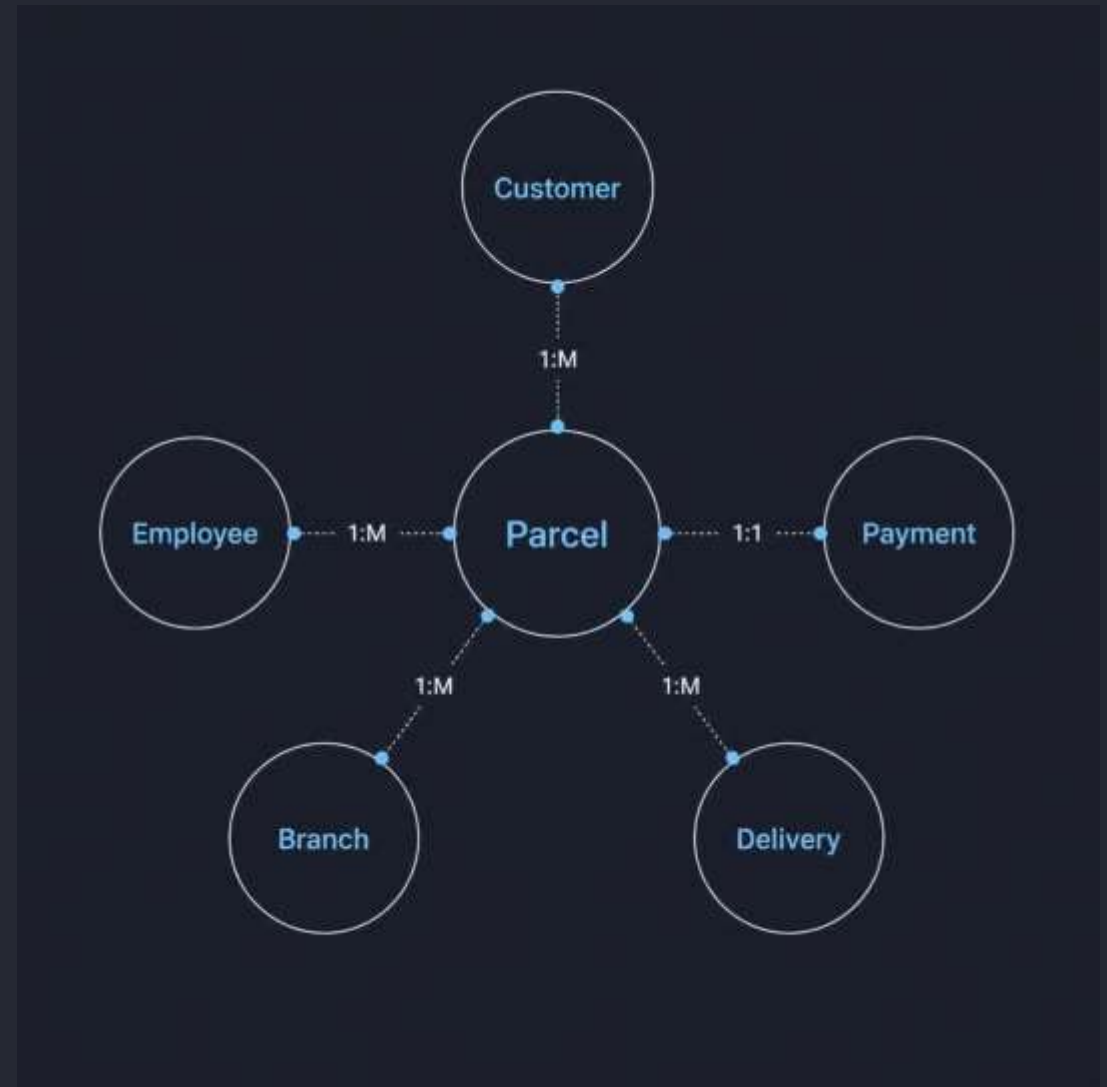
DATABASE DESIGN

ER Diagram

The normalized relational schema captures all core entities and their relationships:

- **Customer** (1:M) → Parcel
- **Employee** (1:M) → Delivery
- **Branch** (1:M) → Employee
- **Parcel** (1:1) → Payment
- **Parcel** (1:M) → Delivery

A fully normalized schema eliminates redundancy and guarantees data integrity across the entire parcel lifecycle.

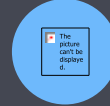


System Modules



Customer Management

Registration, profiles & address management



Parcel Booking

Shipment creation, service selection & cost calculation



Parcel Tracking

Real-time status, location updates & delivery confirmation



Employee Management

Staff profiles, assignments & performance tracking



Payment Management

Invoice generation, payment processing & receipts



Report Generation

Analytics, delivery statistics & revenue reports

TECH STACK

Technology Stack



Frontend

HTML5, CSS3, JavaScript —
responsive UI for all devices

Backend

Java / PHP for server-side logic and
API handling

Database

MySQL (or) Prisma SQL — relational
data management & normalization

Dev Tools

XAMPP, VS Code, Git for development
& version control

Key Advantages & Future Enhancements

ADVANTAGES



Real-Time Tracking

Customers monitor parcels from pickup to final delivery.



Data Security

Encrypted database with role-based access control.



80% Efficiency Gain

Automated workflows dramatically reduce manual effort.



Cost Reduction

Paperless operations and optimized delivery routing.

FUTURE ENHANCEMENTS



GPS Tracking

Real-time location mapping for all delivery vehicles.



Mobile Application

Native iOS & Android apps for customers and couriers.



AI Route Optimization

Machine learning for smarter, faster delivery routes.



Cloud Integration

Scalable cloud deployment for multi-branch operations.



Conclusion

✈ Transforms Operations

Enterprise Parcel Management System automates courier operations end-to-end.

📦 Measurable Benefits

Faster delivery, reduced errors, and a significantly improved customer experience.

🌐 Enterprise-Ready

Scalable, normalized, and ready for deployment across multiple branches.

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